

Calvin Joyce

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Research Interests

Medical robotics and physical human–robot interaction (pHRI): haptic and bilateral teleoperation, robot learning for contact-rich manipulation, clinical safety formalization, continuum/endovascular robots, and humanoids for clinical tasks, grounded in years of hands-on robot hardware design and deployment.

Seeking a medical robotics R&D / New Product Development internship, available January–September 2027.

Education

University of California, San Diego · B.S. & M.S. Mechanical Engineering, GPA 3.8/4.0

B.S. 2024 · M.S. expected June 2027

Research & Work Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego

June 2021 – Present

Advisor: Prof. Michael C. Yip

La Jolla, CA

Grad. Researcher (funded, full-time, 2025–) · R&D Engineer (full-time, 2024–25) · Research Asst. (paid, 2022–24) · Undergrad. Researcher (2021–22)

◦ PHABS (Portable Haptic Assisted Bimanual System)

Project lead; first author, ICRA 2027 submission

- * Invented the first handheld bimanual device to render the internal force between two hands on a shared object, the signal that decides crushing or dropping it, by reacting that force across the operator's hands so the device needs no grounding.
- * Built the bilateral teleoperation framework that renders a dual-arm follower's contact forces as per-hand pinch and signed inter-hand force, and characterized both channels against a reference load cell.
- * Made demonstrations safer in a within-subjects user study (10 participants, 247 trials): pinch feedback cut egg crushes from 33% to 6% and grip force by two-thirds, and the inter-hand channel nearly halved squeeze on a shared-object lift, with no cost in task time.

◦ RIC (Robotic Infant Care) · clinical safety benchmark for robotic infant handling

Project lead; first author, ICRA 2027 submission

- * Introduced CRIB, the first framework scoring robots against clinical safety limits for handling infants, applied to bimanual pickup and CPAP nasal-mask repositioning, two of the NICU's most frequent hands-on tasks.
- * Converted clinical practice into testable acceptance criteria with four NICU clinicians: cervical pitch and head acceleration for pickup; per-site contact force, eye/mouth exclusion zones, and a 15 s PEEP budget for CPAP.
- * Benchmarked direct-human, expert and novice teleoperated, and learned (ACT) handling across 641 instrumented trials on a Unitree G1 and an OpenArm, by building a motion-captured infant manikin and a force-sensing CPAP mask.
- * Showed sustained force regulation is the gap: pickup success ran from 100% (human) to 83% (policy), but on CPAP humans met the pressure bound on just 1.7% of trials, against 61% for expert teleoperation and 32% for the policy.

◦ Humanoid Surgery ([video](#)) · in vivo feasibility of humanoid robots in laparoscopic surgery

Co-author, [Nature 2026](#)

- * Enabled humanoid robots to operate standard medical tools in dry-lab and in vivo porcine surgeries, by designing the tool-mount system (optimized mass/CoM, bench- and sim-validated rigidity) and running mechanical setup for every surgical session.
- * Coordinated a 10-person robot–clinician team through live surgeries, turning surgeon feedback into hardware revisions.

◦ Autonomous Catheter Navigation · early-stage ARPA-H & DARPA project; RL for endovascular robotics

Jul 2026 – Present

- * Building the learned low-level controller that drives a catheter tip to planner-issued waypoints, by training PPO policies with domain randomization in GPU physics, with contact limits anchored to measured tissue perforation forces.
- * Moved training onto real patient anatomy, by building the pipeline from segmented mesh to trainable vessel geometry (centerline and radii extraction, junction trimming, lumen baking) behind a one-command intake for any mesh.
- * Found and fixed a physics fault that had silently disabled the catheter's elasticity, by ruling out device, anatomy, drive, and stiffness with data before tracing it to a containment layer that moved node positions without updating the rod's internal frames.
- * Built a simulated C-arm rendering X-ray views from CT volumes and stood up the training infrastructure behind multi-hour runs; contributed three catheter demos to NVIDIA's open-source Newton engine.

ArtemisAI Labs · Mechanical Engineer, La Jolla, CA

Jan 2026 – Present

- * Made a NICU bedside camera mount clinically deployable, by writing its load, retention, and cleanability requirements (rough-handling loads benchmarked against the IEC 60601-1 instability test; no-disassembly hydrogen-peroxide wipe-down) and redesigning it against them with low-cost, manufacturable, and COTS components.
- * Split the unit into separately specified clamp, arm, and interface subassemblies with acceptance criteria for each, then wrote the assembly protocol and bill of materials so labor could be quoted independently of parts.
- * Reached the bedside: the mount fields the cameras for NOVA, ArtemisAI's Phase II multi-center trial of computer-vision neurological monitoring of newborns at Mount Sinai; qualifying assembly partners to kit and ship units to purchasing hospitals.

Research Projects

• Haptic Shoulder ([paper](#)) · pHRI phantom; co-first author, ICRA 2025; provisional patent

Jan 2024 – June 2024

- * Invented a biomechanically accurate shoulder phantom for safe pHRI testing (provisional patent in progress), by dedicating one motor-encoder module to each anatomical DOF to reproduce joint center, coupled kinematics, and range of motion.
- * Enabled pHRI planning to be tested on physical robots without human subjects, standing in for a patient like a medical phantom.

• ARCSnake V2 (w/ NASA JPL) ([paper](#)) · amphibious screw-propelled snake robot; IROS 2026

June 2021 – Sep 2025

- * Increased propulsion-screw output torque 40%, by raising the torque ratio, aligning the belt drive, and cutting friction (shorter drive stack, relieved rubbing surfaces), validated on a testbed I built.
- * Characterized which screw parameters drive performance across media, by engineering a mobile testbed for constrained axial force measurement, validated by FEA before fabrication (co-first author, ICRA 2023).
- * Designed NASU (Novel Actuating Screw Unit), the first Archimedes-screw locomotion design with a dynamically reconfigurable angle of attack for sand, water, and dirt (first author, ISRR 2024).
- * Built the "Voodoo Doll," a 10-DOF joint-matching teleoperation controller, and led integration for underwater testing.

- Completed a track autonomously, by implementing edge-detection and line-following in Python with OpenCV on embedded Linux; integrated an IMU to detect hydroplaning and trigger corrective steering.

Publications [\(Google Scholar\)](#)

Presented or co-presented at ICRA 2023, ISRR 2024, ICRA 2025, and ICRA 2026. * denotes equal contribution.

1. Z. Liang, N. Thareja, P. Zhang, **C. Joyce**, S. Atar, F. Richter, G. Jacobsen, S. Liu, R. Broderick, and M. C. Yip, "[In vivo feasibility study of humanoid robots in surgery](#)," *Nature*, 2026.
2. **C. Joyce**, J. Honma, T. Tattala, V. Benedicto, and M. C. Yip, "PHABS (Portable Haptic Assisted Bimanual System) for Force-Annotated Bimanual Shared Object Teleoperated Data," submitted to *ICRA 2027*.
3. **C. Joyce**, S. Hong, R. Fellbaum, F. Richter, and M. C. Yip, "How Safe Is Robotic Infant Handling? Benchmarking Learning from Demonstration Against Clinical Thresholds," submitted to *ICRA 2027*.
4. Z. Liang, X. Liang, S. Atar, S. Das, Z. Chiu, P. Zhang, **C. Joyce**, F. Richter, S. Liu, and M. C. Yip, "[LapSurgie: Humanoid Robots Performing Surgery via Teleoperated Handheld Laparoscopy](#)," *ICRA*, 2026.
5. S. Wickenhiser*, E. Peiros*, **C. Joyce**, P. Gavrilov, S. Mukherjee, S. Sylvester, J. Zhou, M. Cheung, J. Lim, F. Richter, and M. C. Yip, "[ARCSnake V2: Mechanical Adaptations For An Amphibious Multi-Domain Screw-Propelled Snake-Like Robot](#)," *IROS*, 2026.
6. **C. Joyce***, E. Peiros*, T. Murugesan*, R. Nguyen, I. Fiorini, R. Galibut, and M. C. Yip, "[Haptic Shoulder for Rendering Biomechanically Accurate Joint Limits for Human-Robot Physical Interactions](#)," *ICRA*, 2025.
7. **C. Joyce**, J. Lim, R. Nguyen, M. Owens, S. Wickenhiser, E. Peiros, F. Richter, and M. C. Yip, "[NASU — Novel Actuating Screw Unit: Origami-Inspired Screw-Based Propulsion on Mobile Ground Robots](#)," *ISRR*, 2024.
8. S. Atar, **C. Joyce***, X. Liang*, F. Richter, W. Ricardo, C. Goldberg, P. Suresh, and M. C. Yip, "[Humanoids in Hospitals: A Technical Study of Humanoid Surrogates for Dexterous Medical Interventions](#)," arXiv:2503.12725, 2025. (co-second author)
9. **C. Joyce***, J. Lim*, E. Peiros, M. Yeoh, P. V. Gavrilov, S. G. Wickenhiser, D. A. Schreiber, F. Richter, and M. C. Yip, "[Mobility Analysis of Screw-Based Locomotion and Propulsion in Various Media](#)," *ICRA*, 2023.

Patents

- Haptic Shoulder: biomechanically accurate shoulder phantom for pHRI testing. Provisional patent, in progress (co-inventor).
- PHABS: handheld bimanual haptic teleoperation device with inter-hand force rendering. Patent in preparation (lead inventor).
- Helping Hand: catheter-tip end-effector for grasping and cutting. Provisional patent, in progress (co-inventor).

Mentorship

- Mentored 7 students on the humanoid surgery project (concluded): CAD/FEA reviews, DFMA, fixture design, and test plans.
- Mentored 2 students each on PHABS and RIC through their ICRA 2027 submissions.
- Mentored 5+ students on ARCSnake V2 through mechanical design reviews, supporting its IROS 2026 paper.
- Coached 20+ mechanical members across Triton Robotics subteams; ran design reviews, fabrication, and assembly for 3 robots.
- Directly mentored 5 members as subteam lead from concept to assembly: requirements, BOM, and manufacturing.

Leadership

Triton Robotics, UC San Diego · Team President / Advisor

June 2020 – June 2023

- Scaled the team 20→80+ members and the proportion of non-male members from 15% to over 35%, by formalizing onboarding and training, DEI partnerships, targeted recruiting, and anonymous feedback loops.
- Led 17 team leads with OKRs, roadmaps, and stand-ups, by founding Business, Marketing, and Project Development teams; produced a formal business plan, sponsorship packet, donation page, and lab fit-out.
- Obtained nationally recognized nonprofit status, leading to \$3K in funding, a corporate sponsorship pipeline, and permanent lab space.

Service & Professional Memberships

- IEEE Robotics and Automation Society (RAS), member since 2023.
- Robotics outreach with Triton Robotics, including events at the UCSD Women's Center.

Skills

- **Medical Robotics & pHRI**: haptics & bilateral teleoperation, catheter/continuum robots, clinical safety thresholds, user studies
- **Software & Control**: Python, C++, MATLAB, PyTorch, ROS 2; imitation learning (ACT), PPO, MPPI; NVIDIA Warp/Newton
- **Hardware**: SolidWorks + FEA, Fusion 360, DFMA, resin/FDM printing, laser cutting; design requirements & acceptance criteria, verification testing, test fixtures, capstan drives, motor drivers, sensors, OptiTrack

Coursework

Robotics; Haptics; Soft Robotics; Intro to Autonomous Vehicles; Linear Control; Signals and Systems; Mechanical Design; Computational Methods in Design; Experimental Techniques; Mechanical Engineering Lab; Ocean Technology Design & Development; Solid Mechanics; Dynamics and Vibrations; Elements of Materials Science; Fluid Mechanics; Heat Transfer; Linear Circuits; MATLAB for Engineering Analysis; Recommender Systems and Web Mining